

BANARAS HINDU UNIVERSITY

B.Voc. Semester II Examination, 2022-23

Paper: BVCA-205 — Skill Training in Python

Time: Three Hours

Maximum Marks: 70

*Instructions: Attempt all sections. Write answers in your own words as far as practicable.***SECTION — A**

Note: Long-answer questions to be answered in about 500 words each. Attempt any two. [12 × 2 = 24]

Question 1. What is a programming language? Explain the features of Python programming language.

Question 2. What do you understand by iteration? What are the various loop structures supported by Python programming language?

Question 3. What is searching? Explain the Binary Search algorithm with a suitable program.

Question 4. What are operators? Explain all the operators supported by Python programming language.

SECTION — B

Note: Attempt any six questions. Answer in about 200 words each. [6 × 6 = 36]

Question 5. Write a Python program to find the sum and average of N natural numbers.

Question 6. Write a Python program to print the Fibonacci series up to N.

Question 7. Write a Python program to find the factorial of a given number using recursion.

Question 8. Write a Python program to check whether a given number is prime or not.

Question 9. Write a Python program to create a list with different types of elements and access elements of the list using loops.

Question 10. Write a Python program to find the largest element in an array.

Question 11. Write a Python program to implement Selection Sort.

Question 12. Write a Python program to perform various operations on strings such as copy, concatenate, and compare.

Question 13. Write a Python program to perform matrix addition.

SECTION — C

Note: Attempt all questions.

[10 × 1 = 10]

Question 14. Answer the following:

(i) Which of the following is the correct extension of the Python file?

- (a) .python
- (b) .pl
- (c) .py
- (d) .p

(ii) What will be the value of the following Python expression? $4 + 3 \% 5$

- (a) 7
- (b) 2
- (c) 4
- (d) 3

(iii) Which keyword is used for functions in Python?

- (a) Function
- (b) def
- (c) Fun
- (d) Define

(iv) What will be the output of the following Python expression?

```
len({"hello", 2, 4, 6})
```

- (a) Error
- (b) 6
- (c) 4
- (d) 3

(v) How is a code block indicated in Python?

- (a) Brackets
- (b) Indentation
- (c) Key
- (d) None of the above

(vi) What will be the output of the following code snippet?

```
example = ["Sunday", "Monday", "Tuesday", "Wednesday"]  
del example[2]  
print(example)
```

- (a) ['Sunday', 'Monday', 'Tuesday', 'Wednesday']
- (b) ['Monday', 'Tuesday', 'Wednesday']
- (c) ['Sunday', 'Monday', 'Wednesday']
- (d) ['Sunday', 'Tuesday', 'Wednesday']

(vii) What will be the output of the following Python program?

```
i = 0  
while i < 5:  
    print(i)  
    i += 1  
    if i == 3:  
        break  
else:  
    print(0)
```

- (a) Error
- (b) 0 1 2 0
- (c) 0 1 2
- (d) None of the mentioned

(viii) What will be the result of the following expression in Python: $2 ** 3 ** 2$?

- (a) 64
- (b) 512
- (c) 169
- (d) None of these

(ix) Which of the following statements are used in Exception Handling in Python?

- (a) try
- (b) except
- (c) finally
- (d) All of the above

(x) Which of the following is a Python tuple?

- (a) [1, 2, 3]
- (b) (1, 2, 3)

(c) {2, 2, 3}

(d) {}